

ISC Class 11 Computer Science – First Terminal Examination (Set 2)

Maximum Marks: 60

Time: 2 hours

All questions are compulsory. Internal marks are indicated against each question.

Section A (20 Marks – Short Answer Questions)

1. (a) State two functions of the operating system. (2)

(b) Represent -45 in 8-bit two's complement form. (2)

(c) Differentiate between ASCII and Unicode. (2)

(d) Write the complement of the Boolean expression: $(X + Y)(X' + Z)$. (2)

(e) Construct the truth table for $(A \oplus B) \cdot (A + B)$. (3)

(f) What is the significance of the 'break' and 'continue' statements in Java? (2)

(g) Explain the scope of variables in Java with respect to block and class. (3)

(h) List any two differences between procedure-oriented programming and object-oriented programming. (2)

Section B (40 Marks – Programming & Application Questions)

2. (a) What is method overloading? Give a suitable example in Java. (4)

(b) Predict the output of the following code snippet: (3)

```
class Demo {  
    void display(int a) { System.out.println(a * a); }  
    void display(double a) { System.out.println(a / 2); }  
    public static void main(String args[]) {  
        Demo d = new Demo();  
        d.display(4);  
        d.display(5.0); }  
}
```

3. (a) Write a recursive method to generate the nth Fibonacci number. (4)

(b) State one advantage and one disadvantage of recursion compared to iteration. (3)

4. (a) Write a Java program to count the number of vowels and consonants in a string. (6)

(b) Write a Java program to reverse a string without using the reverse() function. (6)

5. (a) Write a Java program to accept 10 numbers in an array and display only the prime numbers. (6)

(b) Write a Java program to input a 2×2 matrix and display the sum of its diagonal elements. (6)

6. Consider the following recursive method: (2)

```
int solve(int n) {  
    if(n == 1) return 1;  
    else return n * solve(n-1);  
}
```

Trace the output for solve (4). (2)
Also state the general purpose of this function.